

IoT-enabled emissions monitoring for energy facilities to comply with new environmental regulations.

Energy × Emissions and environmental monitoring × IoT platform · OS129 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
64 is the world moving	43 can we play, can we win	HIGH 0 named in the evidence	€171m observed evidence · per year	NOW budgeted procurement within 1...	L1 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is an IoT-enabled emissions monitoring solution for energy facilities, driven by tightening environmental regulations and the need for robust data management. It targets energy operators who must demonstrate compliance with new rules, such as mandatory factory monitoring and EU ETS benchmark updates. The time is now because regulators are actively tightening rules and agencies are assessing emissions at industrial sites.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€207m €46.4m – €2.6bn	€171m €38.2m – €2.2bn	€2.1m €459k – €26.0m	observed
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting IoT platform	51.7 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 2 declared geographies are outside the European reference data (TH); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

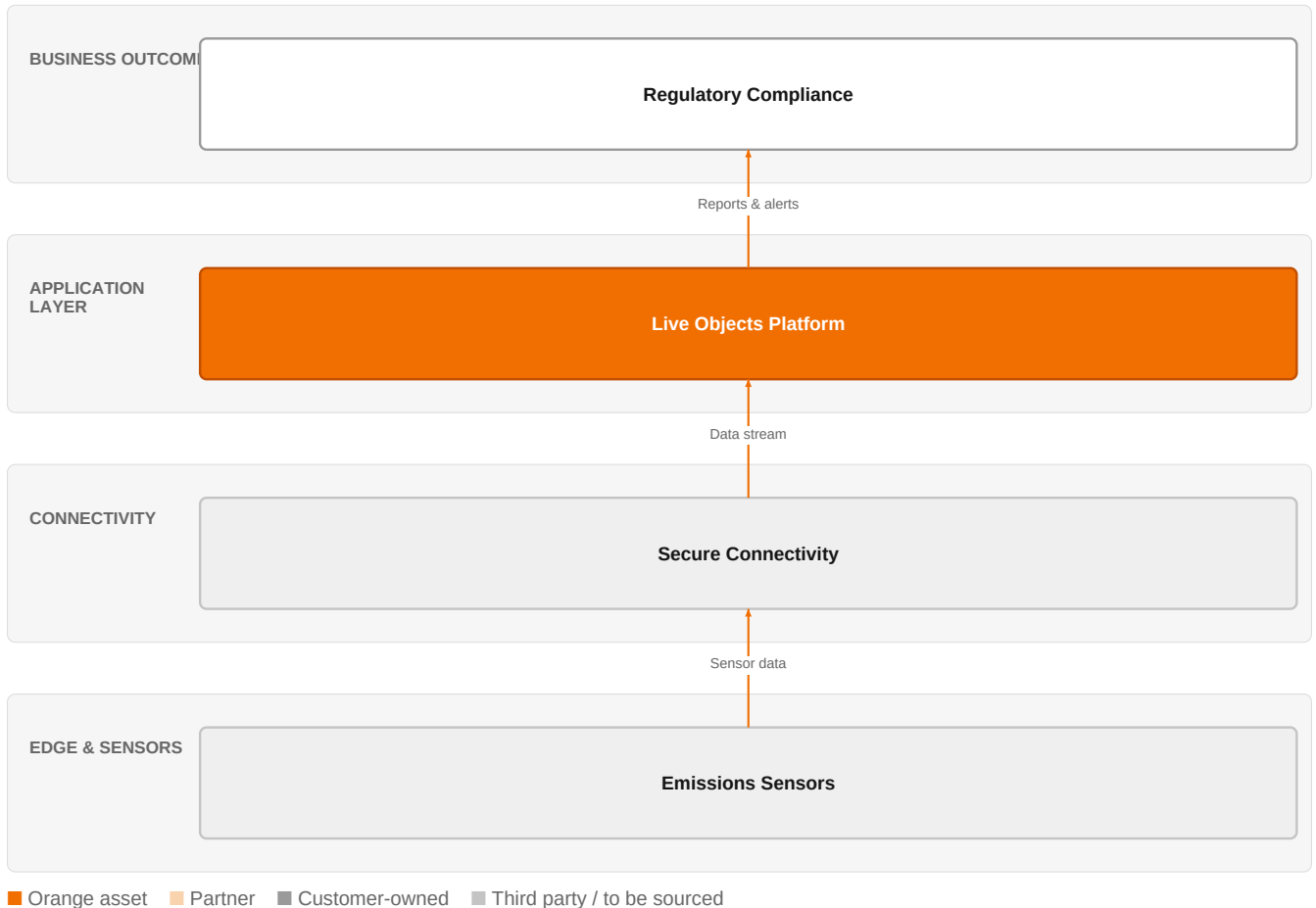
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size. 1 of 2 declared geographies are outside the European reference data (TH); the estimate covers the European footprint only.

The solution, in one picture

IoT Emissions Monitoring Solution



This diagram shows how sensor data flows through secure connectivity into the Live Objects platform, enabling regulatory compliance reporting. Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulators are tightening emissions rules, such as Thailand's mandatory factory monitoring, and the EU ETS benchmark update puts industrial emissions data under sharper scrutiny, increasing the need for robust data management and reporting. Environmental agencies are actively assessing emissions at mining sites, as seen with OGEPA's assessment. These changes create a pressing need for reliable, continuous monitoring and reporting.

Sources: SIG-234C749A6F8C Envirotech Online 2026-06-09; SIG-425F4DF1CB72 Envirotech Online 2026-06-20; SIG-9E90687CF1FE The Guardian Nigeria N 2026-08-16

Who buys, and why now

The buyer is typically the head of environmental compliance or operations at an energy facility, who feels the pain of manual, error-prone emissions reporting and the risk of non-compliance penalties. The trigger is often a new regulation, an audit, or a publicized enforcement action that forces them to upgrade their monitoring capabilities. They need to produce accurate, auditable data to regulators and avoid operational disruptions.

Why it is hot now — every claim bound to a dated source

- Regulators are tightening emissions rules, such as Thailand's mandatory factory monitoring. [SIG-234C749A6F8C]
- EU ETS benchmark updates are increasing the need for robust emissions data management. [SIG-425F4DF1CB72]
- The EU ETS benchmark update puts industrial emissions data under sharper scrutiny, increasing the need for robust data management and reporting. [SIG-425F4DF1CB72]

- Environmental agencies are actively assessing emissions at mining sites, as seen with OGEPA's assessment.
[SIG-9E90687CF1FE]
- Thailand's tightening of air quality rules demonstrates government action to mandate emissions monitoring.
[SIG-234C749A6F8C]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver an end-to-end emissions monitoring solution built on the Live Objects IoT platform, leveraging IoT experts and Orange Group researchers for design and deployment. The engagement would start with a site assessment to identify emission points, then deploy sensors connected to Live Objects for data collection, followed by configuration of dashboards and alerts for real-time monitoring, and finally integration with reporting tools to meet regulatory requirements. The solution would be hosted on certified infrastructure (ISO 27001, SecNumCloud) to ensure data security and sovereignty.

Why Orange can win

Orange can win because it combines a mature IoT platform (Live Objects) with deep expertise in industrial environments and strong security certifications (ISO 27001, SecNumCloud) that are critical for regulated industries. The ability to deliver a sovereign, secure solution differentiates Orange from pure cloud providers. However, the assets are thin on vertical-specific emissions expertise, so Orange must partner with domain specialists to fully address the customer's needs.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitors include AWS, Deutsche Telekom/T-Systems, NTT DATA, Proximus, Telefonica Tech, Telia Company, Verizon Business, and Vodafone Business, all of which sell IoT platforms. AWS is strong in cloud scalability and AI services, while telecom operators like Deutsche Telekom and Vodafone offer connectivity bundles. Telefonica Tech is notably active in the energy sector. The customer will compare on platform maturity, ease of integration, and price, but Orange's security certifications and sovereign cloud options provide a differentiator.

Sources: SIG-234C749A6F8C Envirotech Online 2026-06-09

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	sells IoT platform; competes in OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform; active in Energy	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells IoT platform	structural	—

Vodafone Business	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
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evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently collect and report emissions data to regulators?
- 2 What new regulations or audits are you preparing for in the next 12 months?
- 3 Do you have existing IoT infrastructure or sensors on site?
- 4 Who is responsible for environmental compliance in your organization?
- 5 What is your biggest pain point in emissions monitoring today?
- 6 Are you required to report to the EU ETS or similar schemes?

Objections you will meet

They will say	You can answer
We already have a monitoring system in place.	That's good. Our solution can complement your existing systems by providing a unified platform for data aggregation and advanced analytics, helping you meet stricter reporting requirements with less manual effort.
We are concerned about the cost of implementation.	Cost is a valid concern. Our solution is designed to be modular, allowing you to start with the most critical emission points and scale over time. Additionally, the risk of non-compliance penalties often outweighs the initial investment.
We are worried about data security and sovereignty.	Security is a top priority. Our solution can be deployed on certified infrastructure (ISO 27001, SecNumCloud), ensuring your data is protected and meets sovereignty requirements, which is a key advantage over some competitors.

Risks and unknowns

The main risk is that the customer may already have a preferred IoT platform or may perceive emissions monitoring as a niche add-on rather than a strategic investment. Unknowns include the specific regulatory requirements in the customer's jurisdiction and the availability of compatible sensors. Additionally, the competitive intensity is high, and Orange's lack of vertical-specific emissions expertise could be a disadvantage unless mitigated through partnerships.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for an energy-sector emissions monitoring tender, differentiating with Live Objects' IoT capabilities, ISO 27001 and SecNumCloud certifications, and the backing of Orange Group researchers, and include a reference architecture for EU ETS compliance.
Sales	In a conversation with the environmental compliance lead at a European energy facility, say: 'With the EU ETS benchmarks tightening, how are you currently managing your emissions data? Orange's Live Objects IoT platform can help you automate monitoring and reporting, and we hold ISO 27001 and SecNumCloud certifications to secure that data.'
Strategist / Innovator	Deep-dive the EU ETS benchmark update and Thailand's mandatory factory monitoring to define a reusable IoT emissions-monitoring blueprint that leverages Live Objects and Orange Group researchers, then decide whether to invest in a vertical accelerator for energy facilities.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AC86A5D91D50	2026-06-11	ted.europa.eu	1	France – Pollution tracking and monitoring and rehabilitation – SPCH a Harbonnieres (80) - Etud...
SIG-1F577A1F9D8B	2026-06-09	ted.europa.eu	1	Austria – Sensors – Lieferung von Schwingungssensoren fur eine Carbonseil-Innovationssektion
SIG-D85076BA625E	2026-05-28	ted.europa.eu	1	Lithuania – Methane monitoring – (VPP-680) Gamtinio duju (metano) nuotekiu matavimo, duomenu an...
SIG-B076E0F3616E	2026-05-20	ted.europa.eu	1	Italy – Sensors – DAC.0129.2025-Fornitura e posa di sistemi di monitoraggio per ponti e viadott...
SIG-0205DE7D82FE	2026-04-01	openalex.org	1	ZigBee Based Low Latency IoT and AI Integrated Framework for Real Time Telehealth Monitoring
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-1C6BA6D836AF	2025-12-17	openalex.org	1	FFT-Guided Multi-Window USAD with DTW–Isolation Forest for Reliable Anomaly Detection in Indust...
SIG-AECDD651DA98	2025-12-17	openalex.org	1	Edge-AI integrated secure wireless IoT architecture for real time healthcare monitoring and fed...
SIG-C8282154F431	2025-10-28	openalex.org	1	Lightweight Signal Processing and Edge AI for Real-Time Anomaly Detection in IoT Sensor Networks
SIG-BCCCB520B0E1	2025-10-21	Institute of Electrical a...	1	Design and Integration of IoT-Enabled Health Sensors for Remote Patient Monitoring
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-9E90687CF1FE	2026-08-16	The Guardian Nigeria News	2	OGEPA assesses gaseous emissions, pollutants at Ogun East Mining Site - The Guardian Nigeria Ne...
SIG-4D1BCBC2C864	2026-07-14	Envirotech Online	2	Mercury emissions monitoring for environmental compliance - Envirotech Online
SIG-425F4DF1CB72	2026-06-20	Envirotech Online	2	EU ETS benchmark update puts industrial emissions data under sharper scrutiny - Envirotech Onli...
SIG-234C749A6F8C	2026-06-09	Envirotech Online	2	Thailand tightens air quality rules with mandatory factory emissions monitoring - Envirotech On...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:12+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.